

effective coordination between the base and on-field operators.”

Another interesting product is the HoloSuit, manufactured in India by Mysuru based Kaaya Virtualisation Tech. The motion capture suit is embedded with over 40 sensors to capture the wearer’s movement from head to toe. The wearer’s movements are replayed in the virtual world, and the wearer also feels everything that happens there. For example, the user can operate a machine in the virtual world. The user also gets haptic feedback from the suit. If someone touches, shoves, punches, or shoots the digital twin, the wearer can feel it. It is being deployed by the Indian Army to train soldiers. According to news reports, the company has exported it to the South Korean Army as well, and it is also being used by one of the electricity boards in India to train workers to avoid electrical hazards.

Sports and fitness: Run to win

The metaverse will make sports and fitness much more interesting. We are not talking about immersive views of sports events, or trading of sports NFTs, we are talking about serious training and workout. For those of you who find the gym environment rather dull and boring but are unable to take out enough time to go rough it out in beautiful nature trails, the metaverse could be the perfect escape. Clip on your VR headset and you could be running through your favourite landscape with your friends, as *avatars* though, to keep you company.

Cycle maker Capti has come up with a mind-blowing bike that dynamically switches from an outdoor bike with steering, gliding pedals, virtual shifting and all the works, to a studio cycling bike with fixed handlebars and flywheel! With an immersive 61cm (24-inch) touchscreen, immersive content built

on the Epic Games Unreal Engine, 50 maps, 6100 metres (20,000 feet) of elevation and more than 500km of stunning graphics, you get to go on a fresh trail every day. The patent-pending Smart Ride technology auto-configures steering and flywheel to match the workout selected by you, to deliver an immersive experience. With Terrain Adaptive Resistance, the resistance auto adjusts to the grade in the game. When your *avatar* goes uphill, it gets harder, and then easier when riding downhill, just like riding a bike outdoors. You can do a virtual cycling course, heart rate training, or simply play a fun game that keeps you moving. Fitness is gamified—as you cycle more and more, new levels and content are unlocked.

If running is your thing, then try Zippy, a virtual world that lets you run across major marathon cities like Boston, London, Mumbai, and Tokyo, or through scenic environments like jungle trails and beach runs. The virtual environment syncs with the real-world kinematics of the runner using fitness wearables or treadmill sensors. OliveX’s ultra-immersive running game *Zombies, Run!* is just the right option for fiction lovers. The accompanying narration and music set the pace for your run, which is a mission to save the world. For example, you run faster when you are chased by the zombies, and slow down to pick up supplies! Or, if it is serious athletic practice that you are thinking about, *Ghost Pacer* might be of help. A solution developed by runners, for runners, these AR glasses project a live holographic opponent into your field of view, for you to compete with and train against.

Sports and fitness training by celebrity coaches is also on the rise in the metaverse. BodyCombat is a popular martial arts based group fitness cardio workout that is gaining popularity around the world. They recently teamed up with Odders Labs to transform their workout into a high-

octane VR gaming app on the Quest platform. Led by their famous program directors Rachael Newsham and Dan Cohen, the game pits players against martial arts challenges across a variety of levels, spanning intergalactic deserts and neo-city skylines. Players earn points for their effort and technique, with the program directors boosting their morale and providing cues and tips throughout the workout.

Looks like fitness could become fun after all!

Much more to explore

We saw several more applications of the metaverse in the course of our series so far, ranging from art and entertainment to socialising and marketing.

Socialising: When Facebook changed the company name to Meta, the messaging was clear that the metaverse is the future of socialising. People would no longer be connecting over text chats or even voice or video calls. Instead, they will escape to their favourite virtual hangout in some world in the metaverse, and chill by playing their favourite games or just chatting, as if everyone is sitting together in the same room!

Entertainment: The entertainment industry is also making its presence felt in the metaverse, what with Indian artists like Daler Mehndi and south Indian playback singer Karthik performing in the metaverse. When movies start releasing in the metaverse, it will be a totally transformative experience capable of shaking up the whole industry, because sitting in the living room with a VR headset on, a person can experience the same magic as in a large format theatre with surround sound. The homemade popcorn and coffee would also work out cheaper than in a theatre.

Art: The art industry is benefiting greatly by the metaverse. Digital artists are able to sell their work on the metaverse as NFTs, while art galleries are using the blockchain to